UI design for data collection

Guided image and metadata collection

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The data collection problem

A usable record needs an image and the right context

Separate files

Photos and spreadsheet rows must keep the same identity.

Incomplete context

Missing fields leave uncertainty about what a record describes.

Unnoticed mistakes

An allowed value can still describe the image incorrectly.

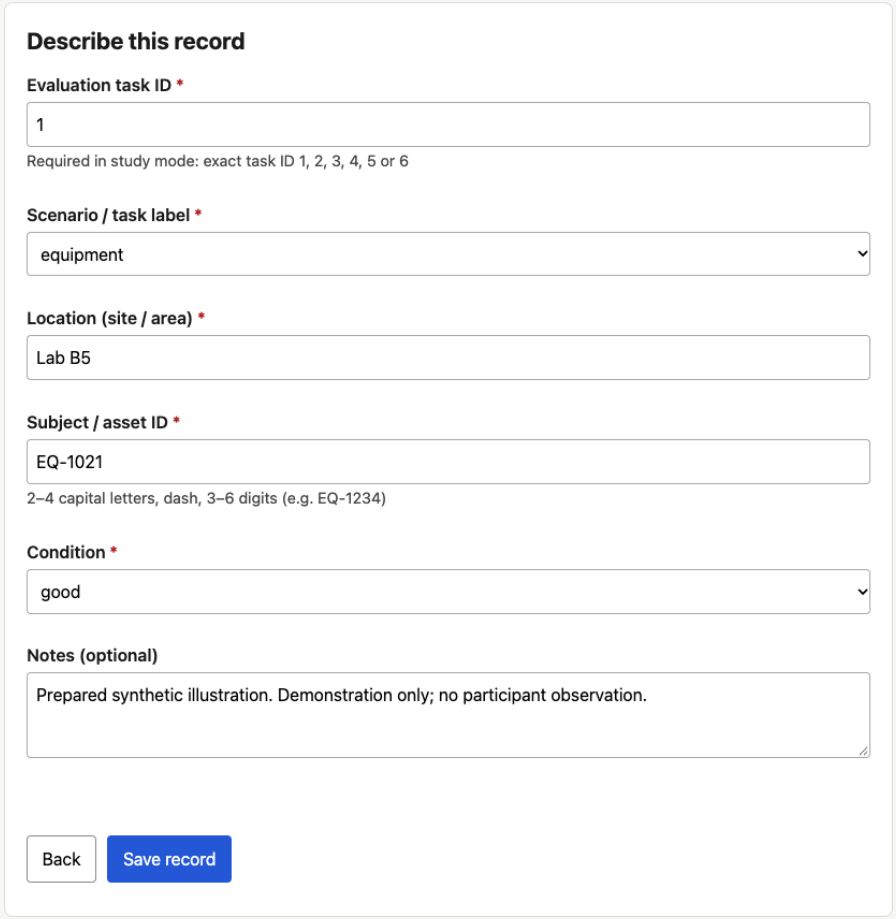
Illustrative workflow risks. Their frequency in an industrial setting has not been measured here.

Project objective and scope

Can a guided workflow reduce missing or incorrect metadata?

What does it cost in completion time?

Current scope
Prepared images and metadata
Primary: remote Guided + Table
Historical: frozen desktop v2
Study payloads omit image bytes



Working scope: image and metadata collection. The human comparison uses remote v3.

Design foundations

Open Data Kit [1]

Established mobile collection architecture and multiple data types.

Web form design [3]

Clear labels, required fields, explicit formats and useful error feedback.

Project contribution

An integrated image workflow, failure recovery, and reproducible checks of the exported records

Human studies inform the design choices. This prototype’s comparative benefit still needs evaluation.

Human evidence behind the design

Form improvements [8]

65 participants
Fewer submission attempts
Two forms completed faster

Format guidance [9]

166 participants
Advance rules reduced errors
and submission attempts

Error location [10]

303 participants
Feedback beside fields outperformed top or bottom messages

Supports a combined set of form improvements

Keep explicit rules visible before entry

Keep corrections beside the affected field

External studies. [8] full text checked. [9–10] abstracts checked. Results depend on the tested tasks and designs.

Validation revision and hypotheses

Remote UI revision

First errors appear on Save
Explicit rules remain visible
Corrections stay beside fields

Planned human comparison

Expect fewer missing or format errors
Measure accuracy, time and SUS
Image–metadata matching stays open

Timing evidence is mixed [13]

Save-time feedback is a testable design choice

research-20261026-r1. Legacy sessions keep their original behavior. Human results remain pending.

The collection workflow

1 Session

2 Import image

3 Enter metadata

4 Inspect, edit and export

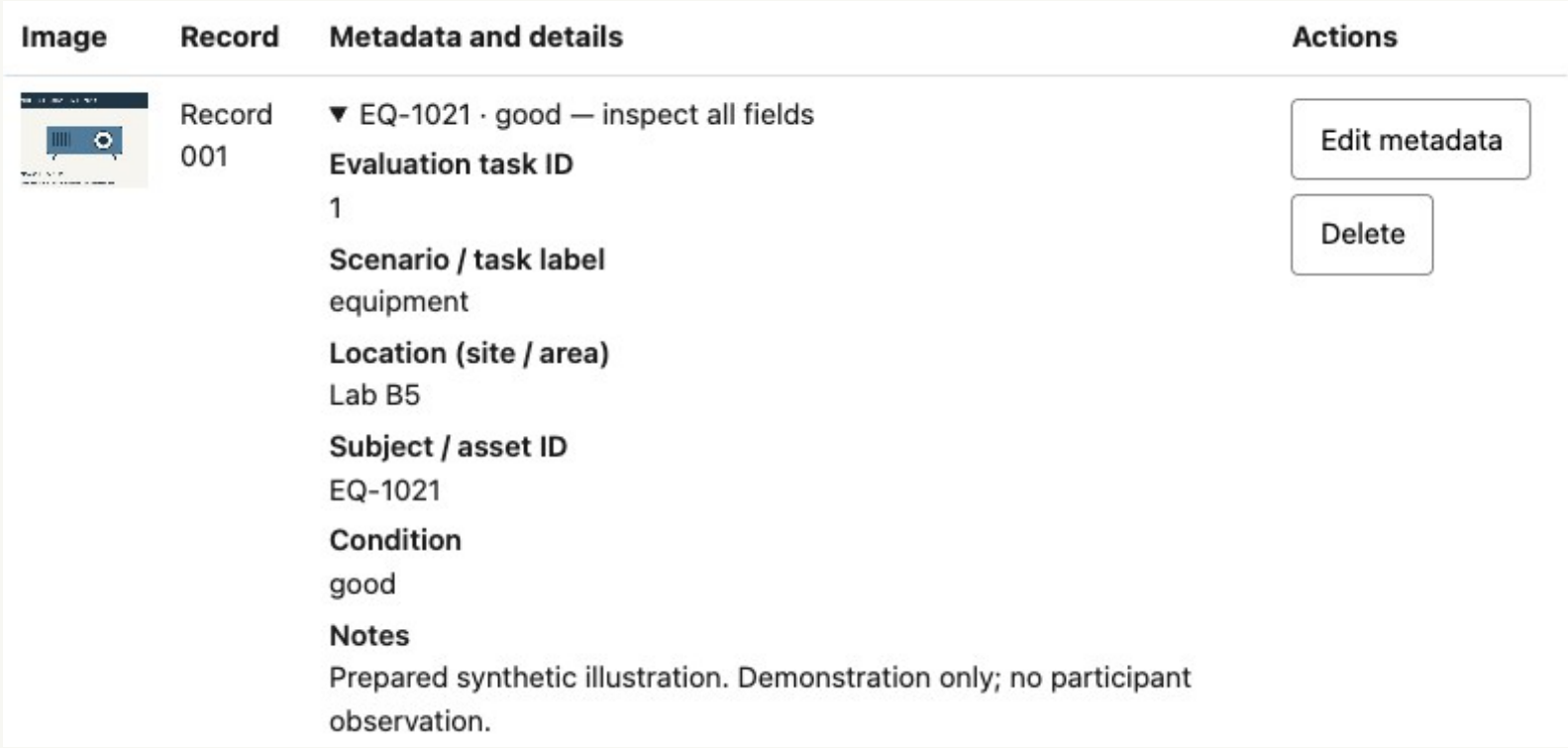
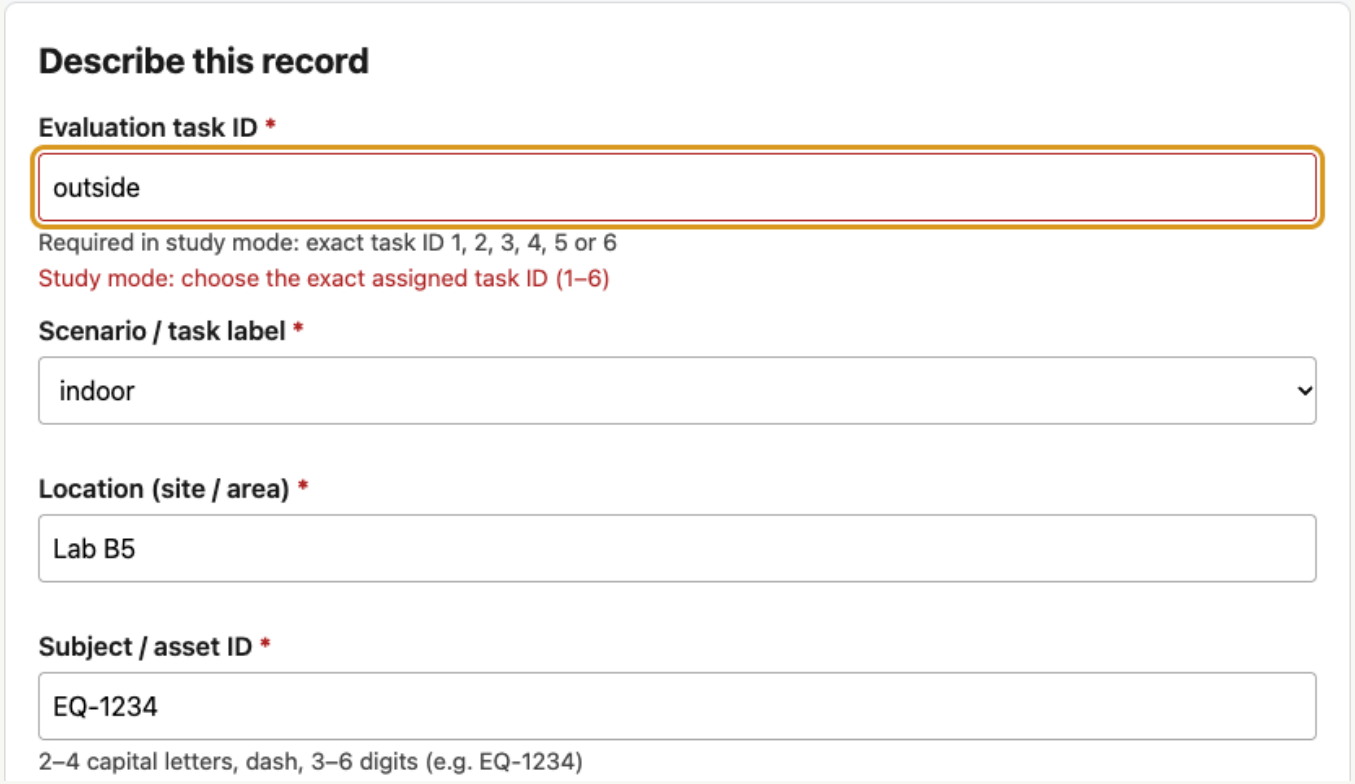


Image	Record	Metadata and details	Actions
	Record 001	▼ EQ-1021 · good — inspect all fields **Evaluation task ID** 1 **Scenario / task label** equipment **Location (site / area)** Lab B5 **Subject / asset ID** EQ-1021 **Condition** good **Notes** Prepared synthetic illustration. Demonstration only; no participant observation.	Edit metadata Delete

[Open remote study]

Frozen v2 review screen, 21 Oct. Remote v3 retains this review and export workflow.

Validation and data accuracy



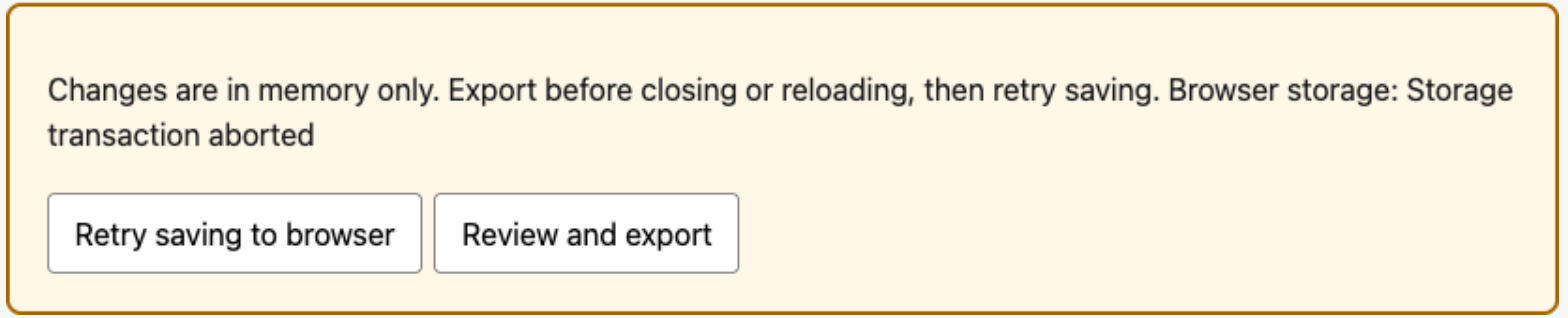
Task identity
Unknown and duplicate IDs block saving

Invalid inputs
Reject malformed values, corrupt images and duplicate task IDs

Allowed values still need a truth check

Validation checks (v2), 21 Oct. Remote v3 also blocked empty guided metadata on 26 Oct.

Recovery from a storage failure



1 record persisted
2 records + 1 draft
in working memory

Recovery ZIP kept both
and the separate draft

Retry and reload
restored the work

The previous commit stays intact

Records and the unfinished draft
remain separately recoverable

Recovery check (v2), 21 Oct: induced transaction abort. Natural quota failure remains untested.

Traceable export

ZIP contents	Purpose
Image files	Retain the collected data
CSV	Spreadsheet text protection; declared policy
JSON manifest	Raw metadata, IDs, hashes and update history

6 of 6 image hashes matched in both checkpoints

v2 ZIP checks, 21 Oct. Remote v3 Chrome ZIP audits also passed on 26 Oct. Hashes do not establish scene truth.

Engineering evaluation

Dated engineering evidence from two separate instruments

Evidence	Observed	Scope
Frozen v2, 21 Oct	82/82 + 42/42	Each Mac/Linux suite. Node: 47/47
Remote v3, 26 Oct	6 + 6 records	Chrome ZIPs opened and audited
AWS readback	S005, seq. 43	15:35:13 HKT. Synthetic, no image bytes

S005 completed both conditions. Both questionnaires were skipped

Agent-run checks only. No human results, SUS score or interface-speed comparison.

Complete metadata can still be wrong

6 tasks × 4 required fields = 24 metadata cells

Checkpoint in one guided run	Complete	Correct
Initial export	24 / 24	23 / 24
After direct metadata edit	24 / 24	24 / 24

An allowed but wrong condition passed validation

One synthetic run, 21 October. The script knew the error; the edit retained the record ID and image.

Remote workflow comparison (v3)

Browser Table

Six rows with raw text cells
Explicit prepared-image selection

Guided workflow

Versioned validation, review and ZIP
Same task cards and image choices

Six tasks each. Alternating order. Two-minute practice per condition

Planned human measures: quality, block time and SUS [4]

Analyze one UI revision at a time. The frozen v2 folder/spreadsheet protocol remains separate.

Current contribution and remaining work

Implemented	**Guided collection and local ZIP export, plus a separate remote study instrument**
Demonstrated	Dated synthetic checks, Chrome ZIP audits and an acknowledged AWS submission
Still to establish	Human usability, device behaviour and fit with the industry workflow

Questions

Evaluation measures

Measure	Definition
Completeness	Filled required cells / all required cells
Accuracy	Correct required cells / all required cells
Completion time	Whole six-task block, including recovery
SUS [4]	Ten-item questionnaire, 0–100 composite

Count missing tasks. Match explicit task IDs rather than row position. Human timing and SUS are pending.

Limitations and AI assistance

Current limits

26 Oct: synthetic desktop Chrome run
Both SUS questionnaires skipped
No human or physical-camera results

AI assistance

Code, tests and synthetic fixtures
Analysis, report and presentation
Disclosed contribution to this work

The current build needs my personal rehearsal

Student responsibility: explain the design, verify the evidence,
and distinguish personal work from assisted work

Selected references

[1] Hartung et al. (2010)

Open Data Kit: Tools to Build Information Services
for Developing Regions. ICTD.

[3] Bargas-Avila et al. (2010)

Simple but Crucial User Interfaces in the World Wide Web:
Introducing 20 Guidelines for Usable Web Form Design. DOI 10.5772/9500.

[4] Brooke

SUS: A quick and dirty usability scale.
Chapter reproduction hosted by AHRQ.

Empirical sources [8–13]: full citations and access limits in the research slide notes and report.

Recovery safeguards

Browser recovery

26 Oct report: reference context reset
Follow-up: restored context verified

Server checkpoint

Serial requests and acknowledged snapshots
Receipt follows a server response

Local recovery and server receipt answer different questions

Job 5 deploys the reviewed safeguards, 26 Oct.
Guided nested scrolling remains a protocol review item.

Hosted synthetic checks include controlled browser faults. Frozen v2 and the comparison protocol remain unchanged.

Remote v3 on AWS

Component	Role in the remote study
Amplify Hosting	Desktop Chrome study pages and prepared assets
This browser	Images and drafts, local ZIP export, private resume receipt
Lambda + private store	API, DynamoDB receipts and S3 metadata snapshots

Study uploads omit image bytes. Local ZIPs retain the images

https://main.d3g93doyp3ywja.amplifyapp.com/

Invitation required. S005 readback confirmed synthetic exclusion and both skipped questionnaires.